

Eya Najlaoui

Applied Mathematics & Modeling Engineer

Data Science • Machine Learning • Deep Learning • Generative AI

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Summary

Applied Mathematics & Modeling Engineer (ENSIT), **ranked 1st in the Reliability & Maintenance specialization**, with a Research Master's in Data Science. Specialized in **Machine Learning**, **Deep Learning** and **Generative AI** (LLMs, RAG, agents), I deliver data projects end to end, from data preparation to deployment. **Seeking a first full-time role** as a Data Scientist, ML Engineer or AI Engineer.

Professional Experience

Talan Tunisia Consulting Feb 2026 – Jun 2026
AI & Data Science Intern (End-of-Studies Internship) *Tunis, Tunisia*

- Designed and built **PrioPath**, a cybersecurity decision-support tool based on a **multi-agent architecture**: automatic detection and **prioritization of multi-CVE vulnerability chains**, with remediation recommendations.
- Built a **Knowledge Graph** (Neo4j) unifying **5,000+ CVEs** from **6 threat-intelligence sources** (NVD, EPSS, CISA KEV, ExploitDB, MITRE ATT&CK, CAPEC).
- Trained **GNN** models (PyTorch Geometric) to classify and prioritize vulnerability chains.
- Built a **Graph RAG** pipeline (graph + LLM) with **LangGraph** and conversational agents for analysis and reporting.
- **Stack**: Python, PyTorch Geometric, LangGraph, LLMs (OpenRouter), Neo4j, PostgreSQL, FastAPI, React.

Five Elements Services (5ES) Jul 2025 – Aug 2025
Data Science Intern *Tunis, Tunisia*

- Developed a **product demand forecasting** model (time series, ML) from internal databases (SQL, ERP).
- Optimized an **LSTM** model (regularization, log transformation): **R^2 improved from 66% to 97%**.
- Deployed a **Streamlit** app to visualize product statistics and forecasts.
- **Stack**: Python, SQL, Scikit-learn, TensorFlow, Pandas, Matplotlib, Seaborn, Streamlit.

Projects

Predictive Maintenance for Aircraft Engines | *Python, Scikit-learn, TensorFlow (LSTM)* Demo

- Classification model (Random Forest, **AUC = 98%**) flagging whether an engine is **healthy or critical**.
- Estimated **Remaining Useful Life** (RUL) from multivariate sensor time series (LSTM, **R^2 = 83%**).

Customer Churn Prediction & Retention Dashboard | *Python, XGBoost, SHAP, Power BI, DAX* Dashboard

- XGBoost model tuned for precision at 80% recall (**ROC-AUC = 0.848**, **82% of churners detected**), with calibrated probabilities converted into **revenue at risk ($139K/month)**.
- **SHAP** explanations and **K-Means** segmentation; 4-page **Power BI dashboard** (DAX, star schema) with a prioritized call list and a **what-if** campaign simulator.

FinQuery: Agentic RAG over Financial Reports | *Python, Qdrant, LLM, Streamlit, Docker* Demo

- AI assistant answering questions on financial reports and **citing the source page** of every figure: hybrid search (embeddings + BM25) with **cross-encoder reranking**.
- **Tool-calling agent** (multi-step search, ratio calculator): **80% correct answers vs. 50%** for a standard RAG pipeline (FinanceBench, LLM-as-a-judge evaluation).

Education

Research Master's in Data Science 2025 – 2026
National Higher School of Engineering of Tunis (ENSIT) *Tunis, Tunisia*

Engineering Degree (Diplôme d'Ingénieur) in Applied Mathematics & Modeling 2023 – 2026
National Higher School of Engineering of Tunis (ENSIT) *Tunis, Tunisia*

- **Ranked 1st in the Reliability & Maintenance specialization**.

Preparatory Cycle in Mathematics & Physics 2021 – 2023
Faculty of Sciences of Monastir (FSM) *Monastir, Tunisia*

Technical Skills

ML / DL: Regression, Classification, Clustering, NLP, Time Series, GNNs – Scikit-learn, TensorFlow, PyTorch
Generative AI: LLMs, LangChain, LangGraph, RAG / Graph RAG, Multi-agent systems, Prompt engineering
Data Science: Data preparation, Feature engineering, Statistics, Visualization (Matplotlib, Seaborn, Power BI)
Languages & Tools: Python, SQL, Pandas, NumPy, Git, Docker, FastAPI, Streamlit
Databases: PostgreSQL, MySQL, Oracle, Neo4j

Certifications

NVIDIA: Deep Learning Fundamentals • **Microsoft**: Azure AI Fundamentals
IBM: Machine Learning • Databases and SQL for Data Science with Python

Languages

Arabic: Native
French: Professional fluency
English: B2